Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Kerstin Thiele	Email	Kerstin.thiele@igb.fraunhofer.de
Institute Name	Fraunhofer-Center for Chemical-Biotechnological Processes CBP			
Address	Am Haupttor Bau 1251, 06237 Leuna, Germany			

2. Technology Overview

Title of Proposed Technology	Scale-Up of Chemical Transformations for Sustainable Application in Collaboration with Korean SMEs			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Knowledge Services		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Testing / Inspection / Analysis Techniques	Chemical Processes	Process Equipment Technologies
Concept of Proposed Technology	The proposed technology focuses on the systematic scale-up of chemical and catalytic reactions from laboratory to pilot scale using Fraunhofer CBP's flexible batch and continuous reactor infrastructure [1]. Emphasis is placed on sustainable process design, including efficient use of raw materials, energy optimization, and minimization of waste streams. The technology enables validation of industrial operating modes under realistic conditions, generation of representative product quantities, and early identification of scale-up risks. Potential uses include the development of bio-based platform chemicals, specialty chemicals, and intermediates for polymers, resins, and fine chemicals.			
Rationale for Joint R&D Project	SMEs often lack access to pilot-scale infrastructure and multidisciplinary expertise required for reliable process scale-up. This joint R&D project combines the innovation potential and proprietary chemistries of Korean SMEs with Fraunhofer CBP's scale-up know-how, engineering capabilities, and pilot plant facilities. The collaboration reduces technical and economic risks, shortens time-to-			

	market, and ensures that developed processes meet industrial safety, quality, and sustainability requirements.
Key Words	1. Scaling of chemical processes 2. Optimization of chemical processes 3. Provision of sample quantities

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3-4)	**End of the Project**	Level (5-6)
R&D Objectives	The main objective is to translate SME-developed chemical processes into production-relevant scale while improving process robustness and sustainability. Specific objectives include optimization of reaction conditions, validation of continuous and batch operation modes, assessment of catalyst lifetime and recyclability, and preparation of process documentation suitable for industrial implementation.			
R&D Target Specifications	include achieving stable pilot-scale operation with reproducible product quality, demonstration of kilogram-scale production campaigns, improvement of space-time yields compared to laboratory benchmarks, reduction of solvent and energy consumption, and quantification of waste streams. Safety and environmental compliance indicators will be integrated as key performance metrics.			
R&D Plan	WP1 – Evaluation Phase (Feasibility and Process Assessment) During the first six months, the chemical process will be systematically evaluated with respect to its technical feasibility and compatibility with existing infrastructure. This includes assessment of process parameters, raw materials, and downstream processing requirements to determine overall implementability. WP2-Laboratory Test Laboratory-scale experiments will be conducted to validate the core chemical reactions and process steps under controlled conditions. Key performance indicators such as yield, selectivity, stability, and reproducibility will be analyzed to define optimal operating windows. WP3-Safety Assessment A comprehensive safety evaluation will be performed, including hazard identification and risk analysis of chemicals, reaction conditions, and equipment. WP4-Adjustment of the Pilot Plant Based on laboratory and safety findings, the existing pilot plant will be technically adapted to meet the specific process requirements. This includes modifications to equipment, control systems, and material flows to ensure process compatibility and operational safety. WP5-Commissioning of the Pilot Plant The adjusted pilot plant will be commissioned through systematic start-up procedures and functional testing. Process control, instrumentation, and operational stability will be verified under defined operating conditions. WP6-Pilot-Scale Trials Pilot-scale trials will be carried out to demonstrate process performance under continuous or batch operation. Data on scalability, efficiency, robustness, and product quality will be collected to assess industrial readiness.			

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WP7-Transfer to SME

The validated process will be transferred to the SME, including documentation and standard operating procedures. This step ensures reproducibility, knowledge transfer, and readiness for industrial implementation and commercialization.

		Year 1				Year 2				Year 3			
Work package		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1	Evaluation												
2	Laboratory test												
3	Safety assessment												
4	Adjustment of the pilot plant												
5	Commissioning of the pilot plant												
6	Pilot-scale trials												
7	Transfer to SME												
Milestones													
1	Process transferable to pilot scale												
2	Successful synthesis at a kilogram scale												

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer CBP will lead process scale-up activities, provide laboratory and pilot-scale infrastructure, and conduct reaction engineering, safety assessments, and process optimization. The institute will supply expert personnel, analytical capabilities, and documentation, and will coordinate pilot trials and data evaluation. Fraunhofer CBP will also support preparation of scale-up guidelines and technology transfer packages.
Expected R&R of Korean SME Partner	The Korean SME partner will contribute proprietary synthesis routes, catalysts, or formulations and define target product specifications and market requirements. The SME will support laboratory validation, provide feedback on process feasibility and economic constraints, and participate in pilot trials and data interpretation. In the final phase, the SME will lead commercialization activities and implementation at industrial scale.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Production of pilot quantities for market testing. • Development of streamlined and optimized chemical processes to enhance resource efficiency. • Identification of process limitations and actionable evaluations for industrial scale adoption.

6. References

[1] https://www.cbp.fraunhofer.de/en/range-of-services/chemical-processes.html

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization